

Falsify

An autonomous research lab, built on Omnigent, that studies when AI agents stop correcting themselves, and audits its own work while doing it.

Hack-Nation 7th Global AI Hackathon · Challenge 3: Agentic Scientific Discovery (Databricks × Omnigent) · solo · github.com/daylorwilliams5/falsify

Challenge & user

End-to-end agentic research: literature → hypothesis → experiment → analysis → adversarial challenge → next decision. **User:** AI-safety and agent-systems researchers who need causal evidence about *why* agent behaviour changes, not more pass/fail benchmarks.

What we built

- **Omnigent research organization:** a PI agent, four pods (literature, design, analysis, adversarial; 8 sub-agents), and an **independent methodology reviewer** that audits every decision.
- **Authority enforced in code, not prompts:**
 - L1: the PI acts alone.
 - L2: needs a reviewer PASS.
 - L3: needs a human approval card in Omnigent.
 - Specs are hash-locked; unknown or missing settings fail closed.
- **Controlled subject environments** (FreightRoute) with immutable trial logs, preregistrations, an append-only timeline, and a hard \$20 API spend cap.

Tools

Omnigent 0.16 (orchestration, policies, approval gates) · Claude (lab agents) · subjects: qwen3:8b via Ollama and Claude Haiku 4.5 via the Anthropic API (structured outputs) · Python (environments, analysis, Clopper-Pearson and bootstrap CIs) · 129 tests.

What worked

- **Executable oversight.** The adversarial pod ran a script that never reads the environment. It passed the preregistered validity bar while acting unsafely in 20/20 trials, so the lab threw out a bar everyone (including the human) had approved.
- **The reviewer runs code, not summaries.** It blocked a run over a silent \$8,000 default in the engineer's build.
- **Honest nulls.** Each null was reported with an upper bound and an explicit claim boundary.
- **The design pod argued the follow-up down from three drafts before spending a cent.**

What was hard

- Instrument validity: a livelocked environment, then an inverted validity bar.
- A small subject model that misread numbers in both directions, so we pivoted to Haiku.
- Omnigent's one-shot mode stopping sessions, and review spirals (fixed with a materiality policy).

Results (Claude Haiku 4.5)

- 1 · Under ambiguity, it checks (exp012, 90 trials, preregistered).**
- Evidence that the current plan was invalid was clear, probabilistic, or conflicting; agents worked alone or in teams of four peers.
- Persisting on the invalid plan: **0/90**.
 - With ambiguous evidence, agents **inspected first in 60/60**. The 95% upper bound on persistence is **4.9%**.
 - Peer pressure was *untested*: the peers never disagreed, which the lab flagged rather than claimed.

2 · Advice, not cost, erodes checking (exp013t, 60 trials).

Inspected before committing	Colleagues silent	Colleagues: "skip it"
Inspection free (1 round)	15/15	11/15
Inspection costly (2 of 6 rounds)	15/15	10/15

- **Advisory majority: −30 pts** of verification (95% CI −13 to −47).
- **Time cost: no detectable effect** (+3 pts, CI −13 to +20).
- Agents that skipped deferred to their colleagues. No agent ever chose the invalid route, so **this design could not produce an unsafe outcome**; the preregistered tripwire flagged this, and no safety bound is reported.
- Manipulation checks passed; the primary measure passed its variance checks.

Social discouragement erodes information-seeking. Whether deference extends to *unsafe* advice is the open question. Caveats: n = 15 per cell, and the advice text names "skipping".

How the time was spent (solo, Oct 3)

morning → 10:40	Idea, pitch, research plan, Omnigent setup
10:40 – 12:00	Manual baseline loop; exp001 (40 trials, qwen3): a floor effect, reported as null
12:00 – 13:00	Omnigent lab goes live; exp009: the instrument fails (livelock)
13:00 – 16:00	Rebuild v2.1; exp010: the lab proves its own validity bar inverted; pivot to Haiku
16:00 – 18:00	exp011 pilot + exp012 main (90 trials): a clean null with bounds
18:00 – 19:30	exp013: 3 design drafts, reviewer BLOCK on an \$8,000 bug, run: −30 pt effect
19:30 →	Write-up, one-pager, videos, submission

21 PI decisions, each independently reviewed (timeline.jsonl) · about \$6 of subject-model API spend.

If we had 24 more hours

- The same advisory majority recommending the *unsafe* route: does deference stay safe?
- Strong prior commitment, the untested third clause.
- Live peers forced to disagree.
- Replication on a second model family.

Data: data/trials/*.jsonl (immutable) · results/*.json · decisions/ · critiques/ · specs/PREREG_*.md